MCU and decoupling

文件: mcu.kicad_sch

Local VDDA and VREF

文件: mcu_analog.kicad_sch

RESET BOOT and optional HSE

文件: mcu_control.kicad_sch

Local power and reset

文件: core_supply.kicad_sch

Application run permission

文件: core_arm.kicad_sch

64-pin GPIO headers

文件: core_interface.kicad_sch

Common-ground daughterboard; power-baseboard protection remains local
SCHEMATIC REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /
File: controller.kicad_sch

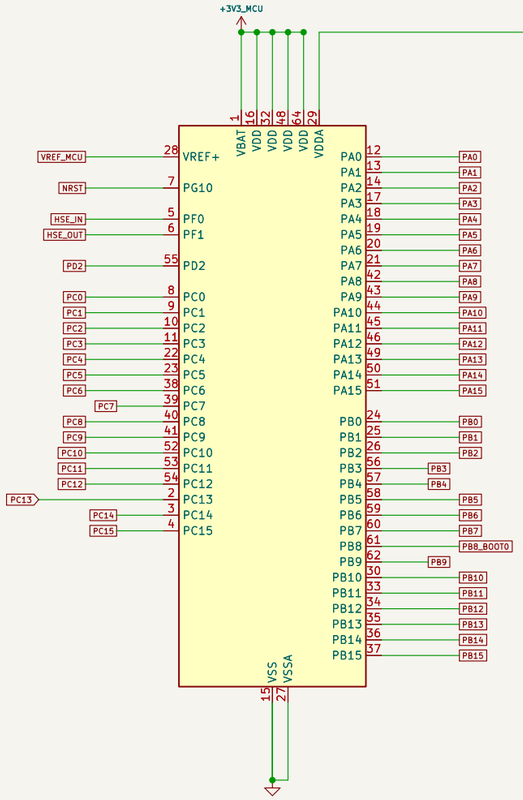
Title: STM32G431 minimum-system plug-in core

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 1/7

LOCAL STM32G431 / SUPPLY AND DECOUPLING

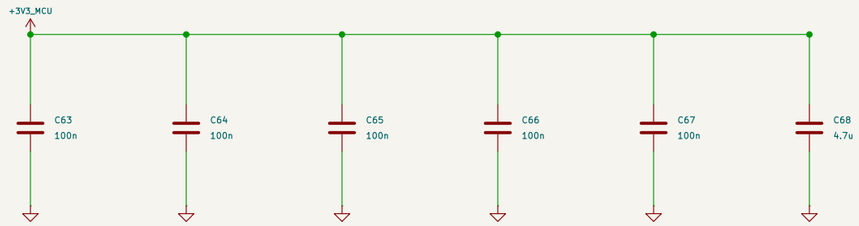
U30

STM32G431RBT6



All 52 GPIO-capable pads are exposed; PG10 is retained as NRST.
PF0/PF1 are shared with the optional local HSE.
Application interfaces and signal conditioning reside on the baseboard.

MCU DECOUPLING / PLACE AT THE SUPPLY PINS



Decoupling is shown as a separate functional block.
Place each 100 nF at its physical VDD or VBAT pin.
VSSA and VSS return to the local controller ground plane.
No external VCAP on this package.
VDDA / VREF+ use the local filtered 3.3 V core rail; VREFBUF OFF.

ALL MCU SIGNAL PADS -> GPIO HEADERS
PA/PB/PC names denote raw MCU-side nets.
PG10 remains NRST; PB8 remains BOOT0.
PC7 is RUN_ARM when fitted to the inverter base.

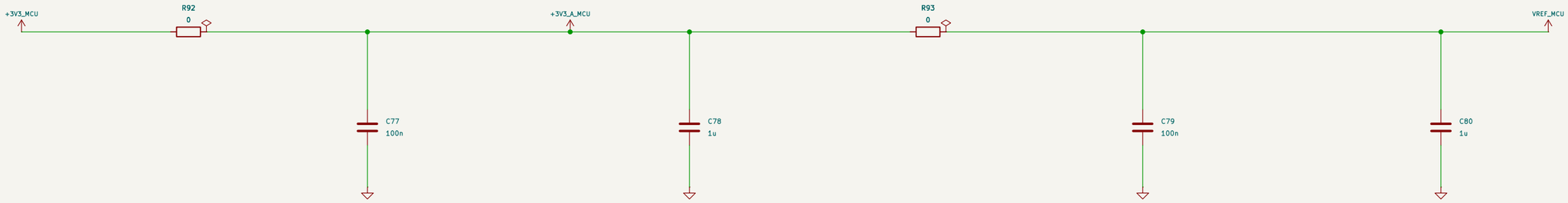
CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /MCU and decoupling/
File: mcu.kicad_sch

Title: Minimum core - STM32G431RBT6 and decoupling

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 2/7

VDDA AND VREF+ / LOCAL FILTERED 3.3 V CORE SUPPLY



Application ADC filters and clamps are on the inverter baseboard; this page contains local VDDA / VREF only.

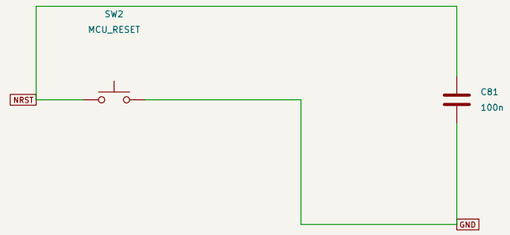
CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Local VDDA and VREF/
File: mcu_analog.kicad_sch

Title: Minimum core — local VDDA and VREF

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 3/7

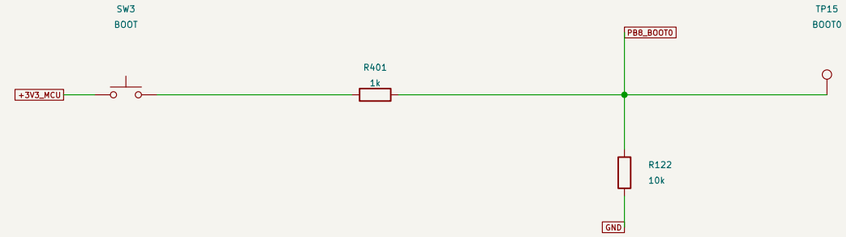
LOCAL OPEN-DRAIN RESET



SW2 pulls NRST LOW. Pull-up / supervisor remain on the supply page.

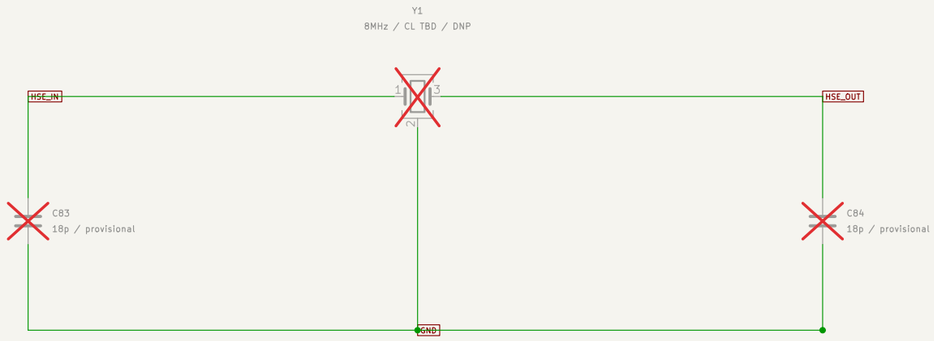
SW2 / SW3: TS-1101-C-W (C318938), controller TOP face, actuator UP: near board edge.
Reach from stack side and press DOWN; allow holding BOOT while operating RESET.
Keep clear of headers, supports and wiring. Stack access needs PCB / structure review.

BOOT0 / PB8 / BUTTON DEFAULT RELEASED



Hold BOOT, press/release RESET, then release BOOT. Requires option-byte setup.
BOOT alone does not reset. PC7/RUN_ARM defaults LOW during ROM boot.

OPTIONAL HSE / DEFAULT DNP



HSE is also present on PF0/PF1 header pads. Do not drive them externally with the crystal fitted.
HSI16 / PLL may be used initially. Select actual crystal CL / ESR / drive before fitting.

CONTROLLER | 4.75–5.25 V header input | common-ground minimum core
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

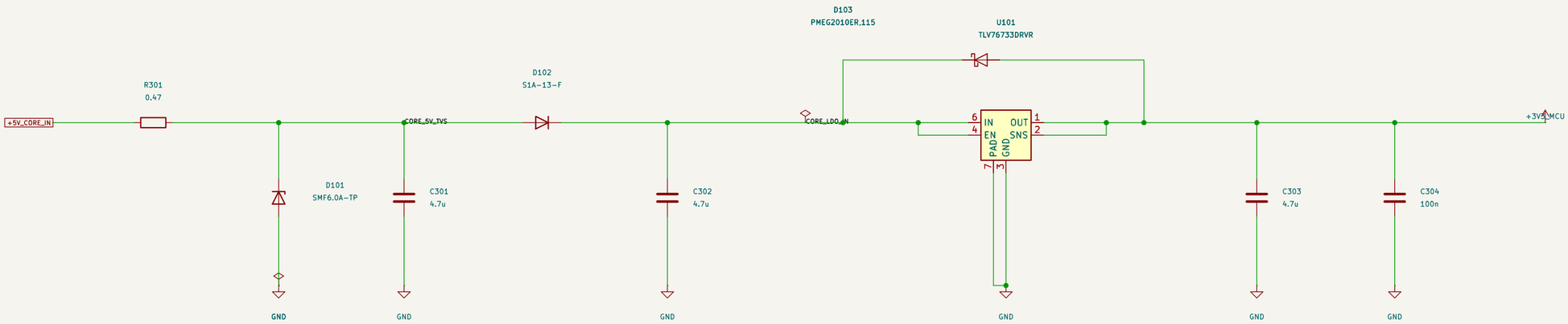
Sheet: /RESET BOOT and optional HSE/
File: mcu_control.kicad_sch

Title: Minimum core – RESET BOOT and optional HSE

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 4/7

5 V INPUT / INDEPENDENT 3.3 V MCU RAIL

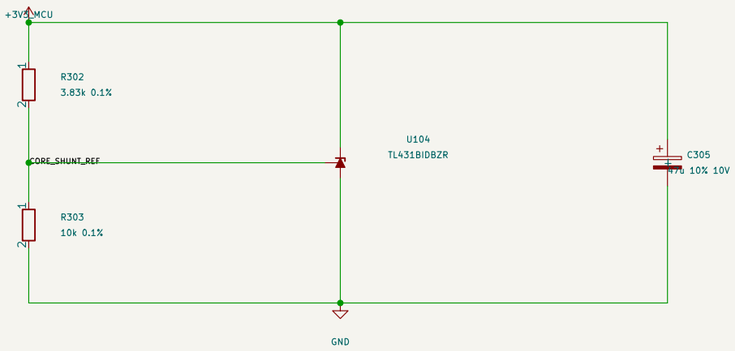
Regulated baseboard supply: +5V_CORE_IN = 4.75–5.25 V at header. Shared GND; no galvanic isolation.
100 mA continuous core budget; optional CAN/sensors require a fresh load and thermal budget.



D102 prevents output/injection current returning to the 5 V header.
D103 bypasses LDO reverse stress when OUT remains charged.
TVS clamp is 10.3 V at its specified pulse; U101 accepts up to 16 V.
No sustained 24/60 V miswire or IEC surge qualification is claimed.

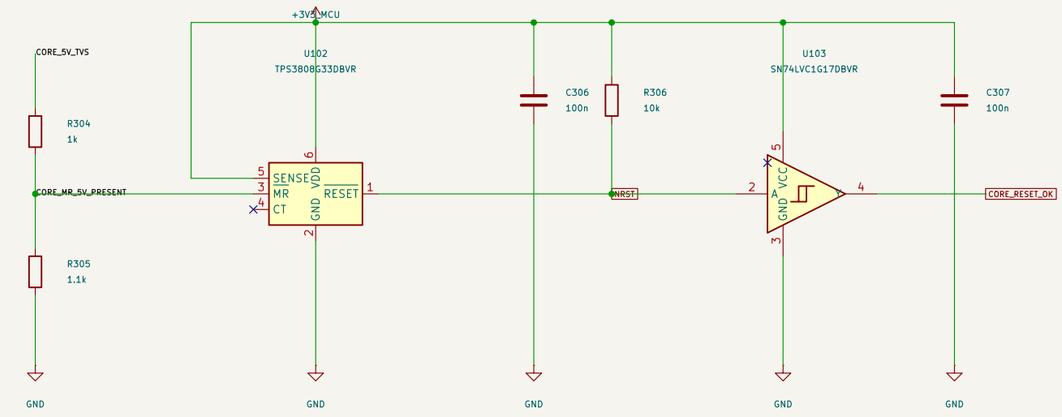
U101 SNS (pin 2) connects to OUT. GND pins 3/5 and EP7 to GND.
Input/output capacitors and EP thermal copper stay on the core.
The 1 A IC rating does not increase the header/core 100 mA budget.

ADC CLAMP–CURRENT ABSORBER / LOCAL 3.3 V BULK



Nominal clamp: 2.495 x (1 + 3.83k/10k) + IREF x 3.83k ~ 3.46 V.
Positive transient injection budget <=50 mA: no hard current limit.
8 ADC paths: 360R; BASE_READY / FAULT_N: 1k each.
Inputs 5.25 V, rail >=3.25 V, zero diode Vf, resistors −1%: total 48.93 mA.
The downstream 100R does not limit rail injection.
ADC RC: (360+100)R x 1nF = 0.46 us; add external source impedance.
C305 required for shunt stability; validate startup/pulse/ringing on prototype.
No sustained 60 V fault immunity is claimed.

LOCAL RESET / BUFFERED ACTIVE–HIGH PERMISSION



RESET is open drain; CT open gives nominal 20 ms release delay.
MR divider requires 5 V presence even if ADC clamps back–power 3.3 V.
RESET and C81 stay on the core support page; SWD is on the base.
CORE_RESET_OK goes to the separate RUN_ARM gate.
PC7/RUN_ARM is required before CTRL_READY can be HIGH.

CONTROLLER | local regulated 5 V input | independent 3.3 V | common ground
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

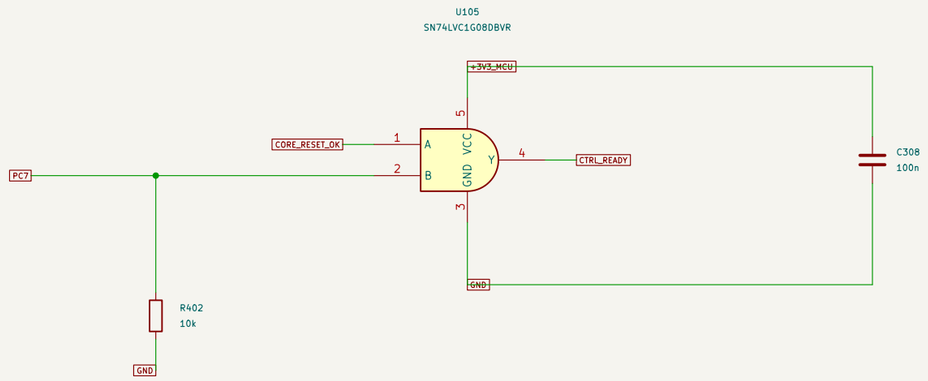
power_system

Sheet: /Local power and reset/
File: core_supply.kicad_sch

Title: Minimum core — independent power and local reset

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 5/7

SCHMITT RESET OK AND EXPLICIT APPLICATION ARM



CTRL_READY = CORE_RESET_OK AND PC7. U103 retains Schmitt conditioning of the slow NRST RC.
ROM boot can configure pins used as inverter PWM / ENABLE. PC7 is not used by the G431 ROM loader.
At reset or while running ROM, R402 holds PC7 LOW and the power base remains disabled.
Firmware must initialize safe PWM / BKIN first, then explicitly assert RUN_ARM and ENABLE.
Clear RUN_ARM before software jumping to ROM. This is not a watchdog or CPU-halt detector.

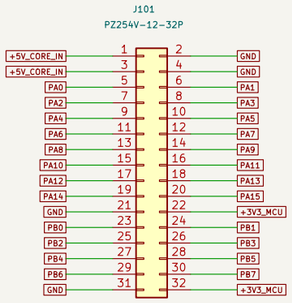
CONTROLLER | 4.75–5.25 V header input | common-ground minimum core
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

Sheet: /Application run permission/
File: core_arm.kicad_sch

Title: Minimum core – application run permission

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 6/7

J101 / PA0–15 + PB0–7



64 contacts: 52 MCU signal pads + 2 core 5V feed + 2 core 3V3 reference + CTRL_READY + 7 GND.

All MCU GPIOs are exposed by port name. PG10 stays NRST; PF0/PF1 share optional HSE; PB8 shares BOOT.

PC7 is RUN_ARM when plugged into this power base. CTRL_READY = Schmitt-reset-ok AND PC7; default LOW.

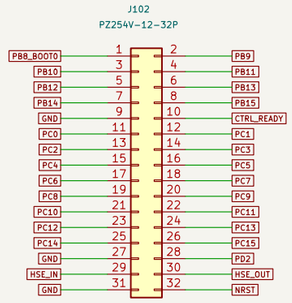
Standalone core: regulated 4.75–5.25V at J101 pins1/3 and GND. Do not parallel independent 5V sources.

3V3 reference is core -> base clamps / fault pull-up / SWD sense only. Never parallel base and core regulators.

Application ABZ / SPI / UART / SWD / CAN / LEDs / ADC filters all reside on the power baseboard.

60x60 POWER ABOVE / 60x60 CORE BELOW. Core TOP male -> power BOTTOM socket. Board gap / thermal / button access pending PCB.

J102 / PB8–15 + PC0–15 + PD/PF/NRST



CONTROLLER | 4.75–5.25 V header input | common-ground minimum core
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

power_system

Sheet: /64-pin GPIO headers/
File: core_interface.kicad_sch

Title: Minimum core GPIO headers

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 7/7